Tolerability and optimal therapeutic dosage of clomipramine for premature ejaculation: A systematic review and meta-analysis

(Supplementary Material)

Supplementary Material

Table S1. Search strategy

Table S2. Risk of bias

Table S3. Certainty assessment by GRADE system

Table S4. Certainty assessment by GRADE system

Figure S1. Small study effect test of intra-vaginal ejaculatory latency time (IELT)

Figure S2. Small study effect test of overall adverse events

Figure S3. Small study effect test of adverse events of nervous system

Figure S4. Forest plot of adverse events of respiratory system

Figure S5. Forest plot of adverse events of gastrointestinal system

Figure S6. Forest plot of adverse events of cardiovascular system

Figure S7. Forest plot of adverse events of renal and urinary system

Figure S8. Forest plot of adverse events of reproductive system

Figure S9. Forest plot of adverse events of dermatological system

Figure S10. Meta-regression analysis of IELT and clomipramine dosage

Figure S11. Meta-regression analysis of adverse events and clomipramine dosage

Supplementary Material

Figure S12. Forest plot of success rate

Figure S13. Small study effect test of success rate

Figure S14. Forest plot of satisfaction rate

Figure S15. Small study effect test of satisfaction rate

Table S1
Search strategy

Table S1
Search strategy

Final syntax in Embase:

('premature ejaculation'/exp OR 'ejaculatio praecox' OR 'ejaculatio precox') AND ('clomipramine'/exp OR '2 chloro 5 (3 dimethylaminopropyl) 10, 11 dihydrodibenz [6, f] azepine' OR '3 chloro 10, 11 dihydro 5 (3 dimethylaminopropyl) 5h dibenz [b, f] azepine' OR '3 chloro 5 (3 dimethylaminopropyl) 10, 11 dihydro (5h) dibenzo (b, f) azepine' OR '3 chloro 5 [(3 dimethylamin) propyl] 10, 11 dihydro 5h dibenz [b, f] azepine' OR '3 chloro 5 [3 (dimethylamino) propyl] 10, 11 dihydro 5h dibenz (b, f) azepine' OR '3 chloroimipramine' OR '5 (gamma dimethylaminopropyl) 3 chloroiminodibenzyl' OR 'anafranil' OR 'anafranil 25' OR 'anafranil retard' OR 'anafranil sr' OR 'anafranilin' OR 'anafranyl' OR 'chlomipramine' OR 'chlorimipramine' OR 'chlorimipramine hydrochloride' OR 'chloroimipramine' OR 'clofranil' OR 'clomicalm' OR 'clomipramin' OR 'clomipramine' OR 'clomipramine chlorhydrate' OR 'clomipramine hydrochloride' OR 'clopress' OR 'domipramine' OR 'equinorm' OR 'g 34586' OR 'g34586' OR 'gromin' OR 'hydiphen' OR 'monochlor imipramine' OR 'monochlorimipramine' OR 'monochloroimipramine' OR 'placil' OR 'zoiral')

Final syntax in PubMed:

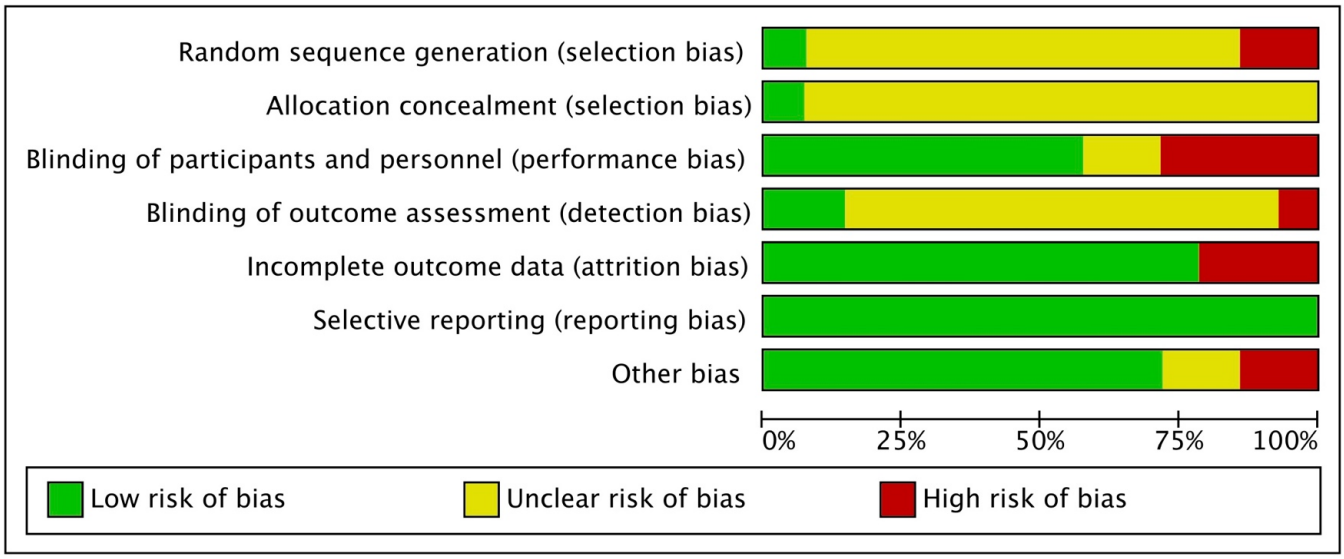
(premature ejaculation OR early ejaculation OR rapid ejaculation OR rapid climax OR early climax OR premature climax OR ejaculatio praecox) AND (Tricyclic antidepressants OR Tricyclic antidepressant OR TCAs OR clomipramine OR Anafranil OR Clomicalm OR 3-Chloroimipramine OR G-34586)

Final syntax in Web of Science:

Topic: (premature ejaculation OR early ejaculation OR rapid ejaculation OR rapid climax OR early climax OR premature climax OR ejaculatio praecox) AND Topic: (Tricyclic antidepressants OR Tricyclic antidepressant OR TCAs OR clomipramine OR Anafranil OR Clomicalm OR 3-Chloroimipramine OR G-34586)

Table S2
Risk of bias

Table S2
Risk of bias summary



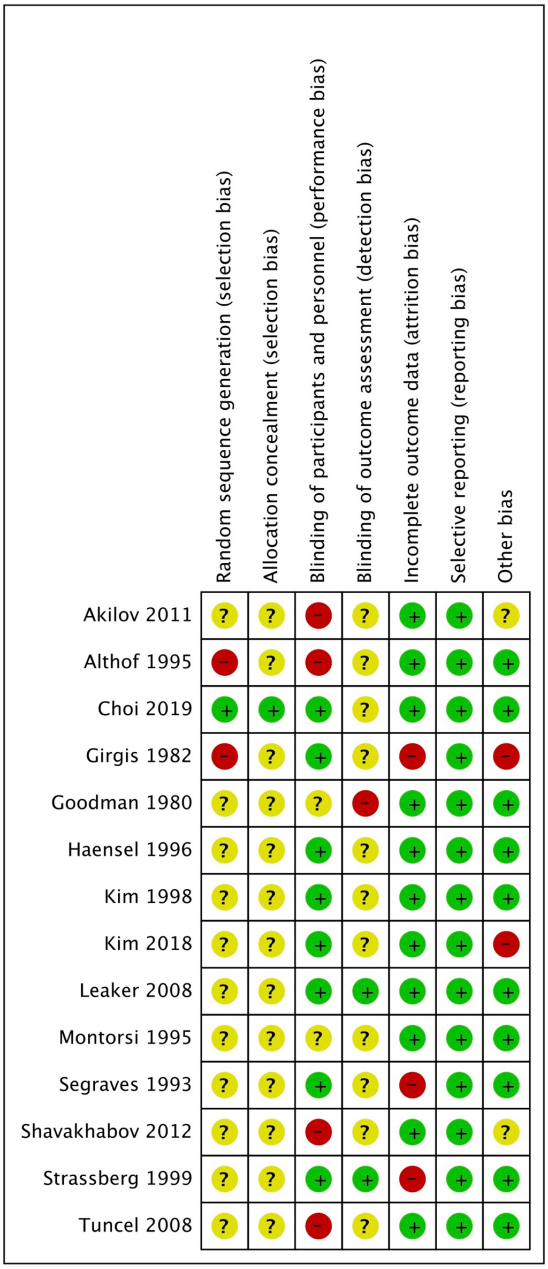


Table S3 to S4
Certainty assessment by GRADE system

Table S3
Certainty assessment by GRADE system

Certainty assessment							
Participants (studies) Follow up	Risk of bias	Inconsistency	Imprecision	Indirectness	Publication bias	Overall certainty of evidence	Effect size (95% CI)
Intravaginal ejaculatory latency time (IELT)							
445 (6 RCTs)	serious[a]	serious[b]	serious[c]	not serious	not serious	⊕⊕◯◯[d] LOW	MD 1.47 (0.73 - 2.21)
Overall adverse events							
514 (7 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕⊕◯[e] MODERATE	RR 3.52 (2.44 - 5.09)
Adverse events of nervous system							
453 (8 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕⊕◯[e] MODERATE	RR 4.07 (2.41 - 6.86)
Adverse events of respiratory system							
430 (6 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	RR 3.80 (1.60 - 9.06)
Adverse events of gastrointestinal system							
289 (4 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	RR 2.11 (0.67 - 6.65)
Adverse events of cardiovascular system							
199 (2 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	RR 2.73 (0.31 - 24.17)

Footnotes:

a. More than 25% of included studies with at lest one high risk in RoB
b. Heterogeneity more than 50%
c. Patient number fewer than 2000
d. Upgrade due to IELT MD 1.47 was clinically and statistically significant
e. Upgrade due to large effect size (RR and lower value of 95% CI more than 2)

Table S3
Certainty assessment by GRADE system

Certainty assessment							
Participants (studies) Follow up	Risk of bias	Inconsistency	Imprecision	Indirectness	Publication bias	Overall certainty of evidence	Effect size (95% CI)
Adverse events of renal and urinary system							
100 (1 RCT)	very serious[a]	not serious	serious[c]	not serious	not serious	⊕◯◯◯ VERY LOW	RR 1.49 (0.06 - 35.58)
Adverse events of reproductive system							
199 (2 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	RR 2.73 (0.31 - 24.17)
Adverse events of dermatological system							
269 (3 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	RR 0.49 (0.11 - 2.22)
Success rate							
57 (2 RCTs)	very serious[a]	not serious	serious[c]	not serious	not serious	⊕◯◯◯ VERY LOW	RR 2.00 (0.99 - 4.05)
Satisfaction rate							
1254 (4 RCTs)	serious[a]	not serious	serious[c]	not serious	not serious	⊕⊕◯◯ LOW	SMD 0.40 (0.09 - 0.70)

Footnotes:

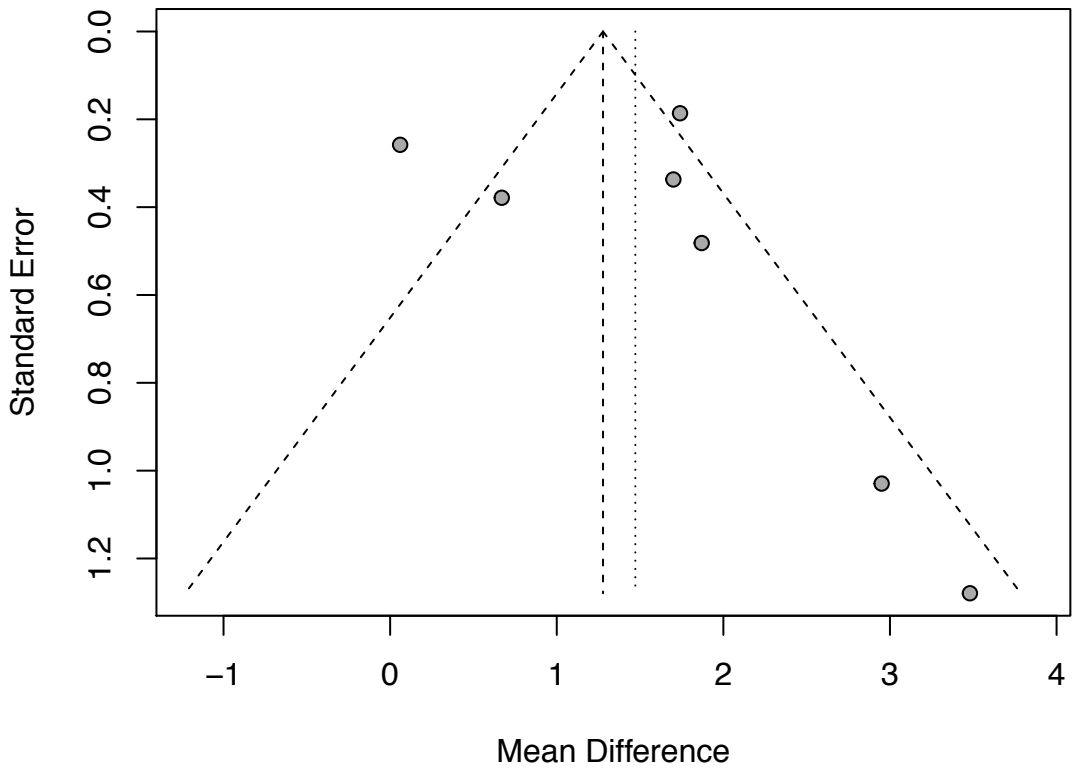
a. More than 25% of included studies with at lest one high risk in RoB
b. Heterogeneity more than 50%
c. Patient number fewer than 2000

Figure S1 to S3

Small study effect test of IELT, overall adverse events, and adverse events of nervous system

Figure S1

Small study effect test of intra-vaginal ejaculatory latency time (IELT)

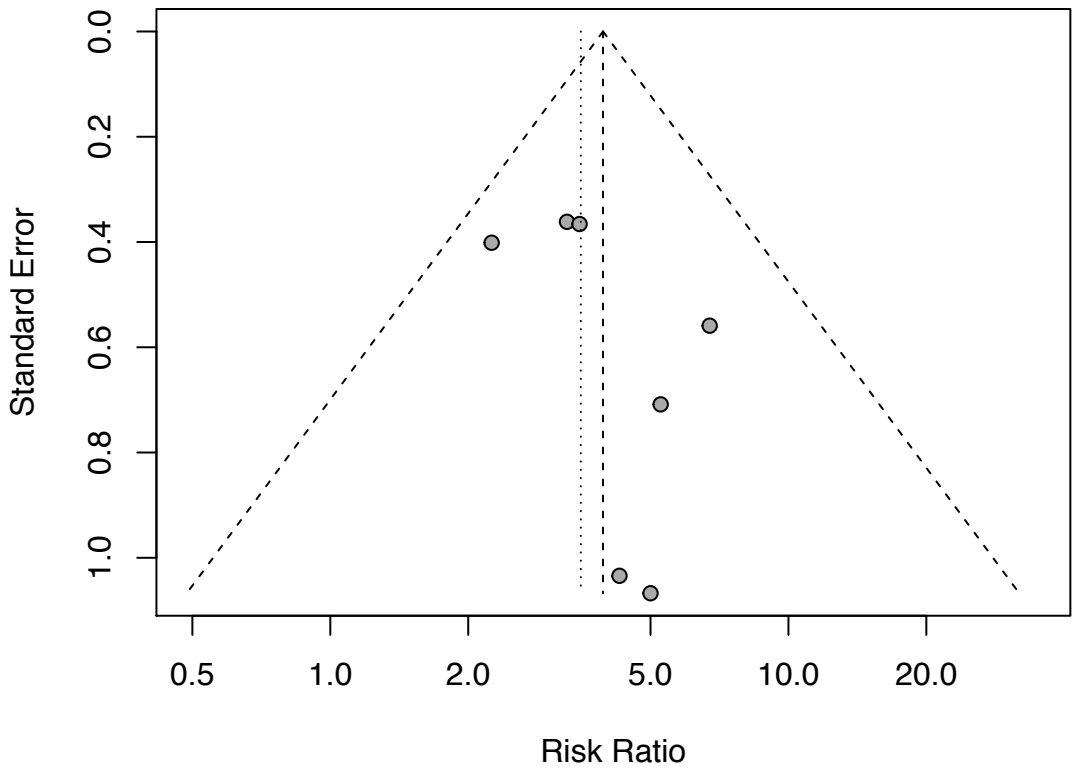


Egger's test intercept

Intercept	95% confidence interval	t-value	P-value
1.189	-2.927 – 5.305	0.562	0.59808

Figure S2

Small study effect test of overall adverse events

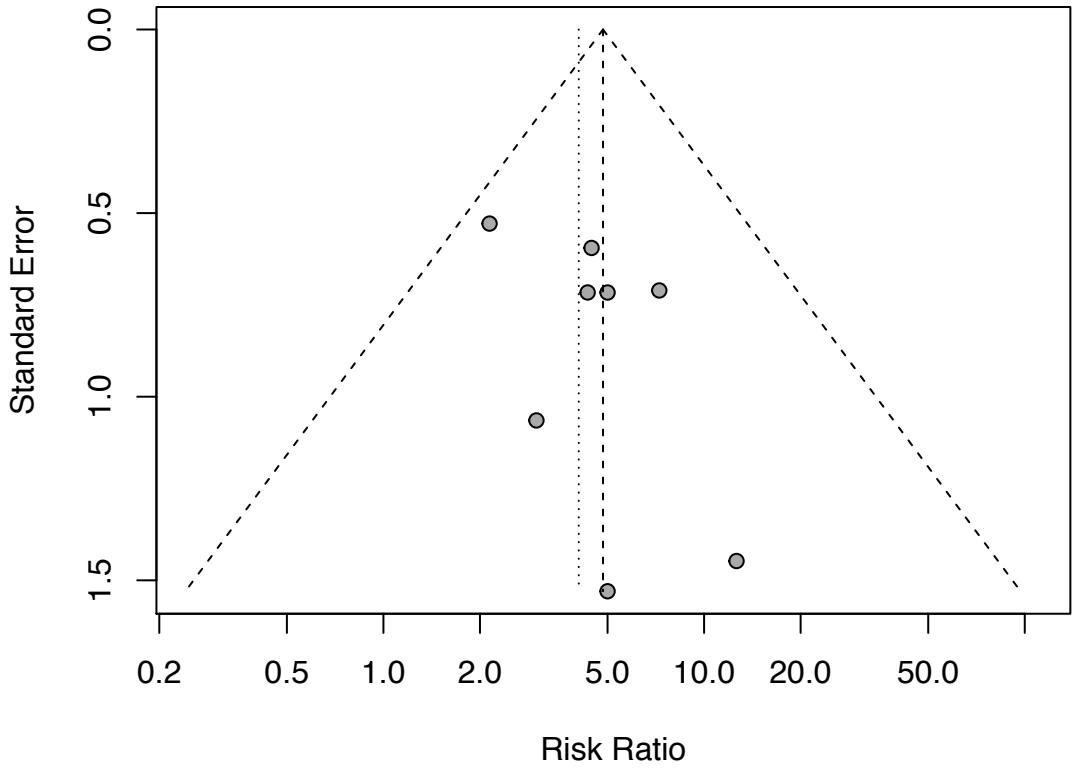


Egger's test intercept

Intercept	95% confidence interval	t-value	P-value
0.965	-0.407 – 2.337	1.414	0.21648

Figure S3

Small study effect test of adverse events of nervous system



Egger's test intercept

Intercept	95% confidence interval	t-value	P-value
0.943	-0.429 – 2.315	1.392	0.21323

Figure S4 to S9

Forest plot and small study effect test of adverse events of other systems

Figure S4
Forest plot of adverse events of respiratory system

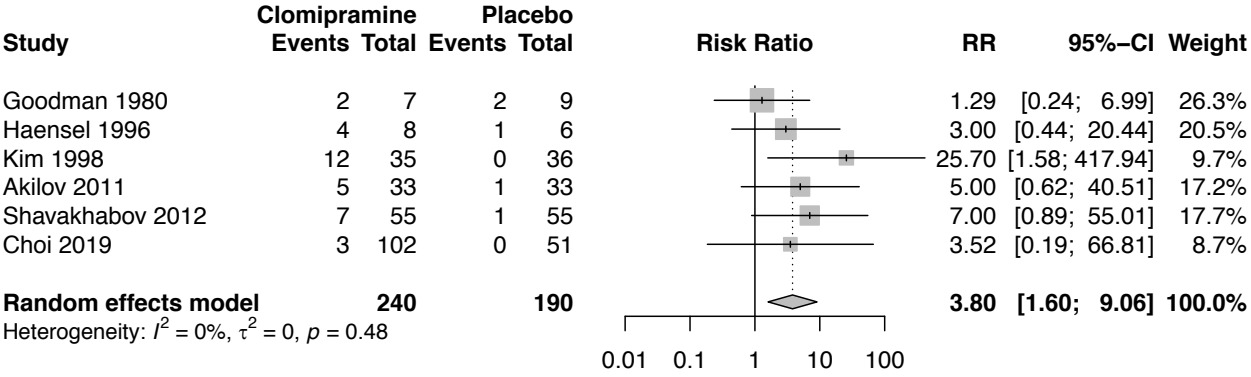


Figure S5
Forest plot of adverse events of gastrointestinal system

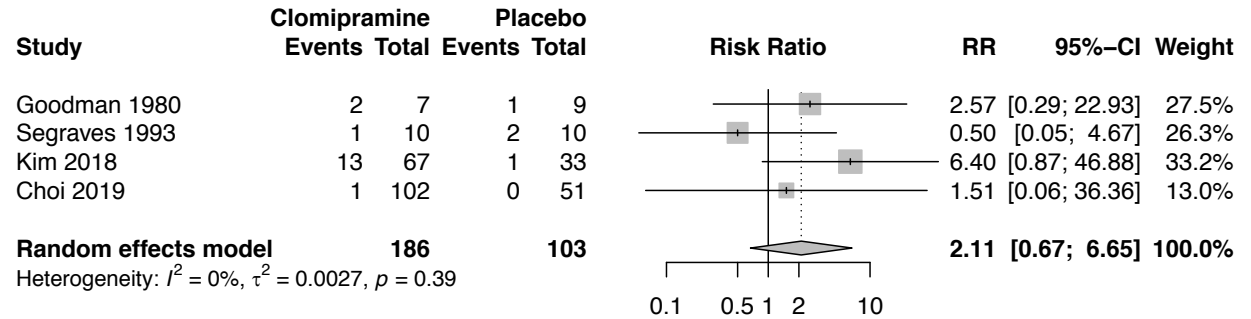


Figure S6
Forest plot of adverse events of cardiovascular system

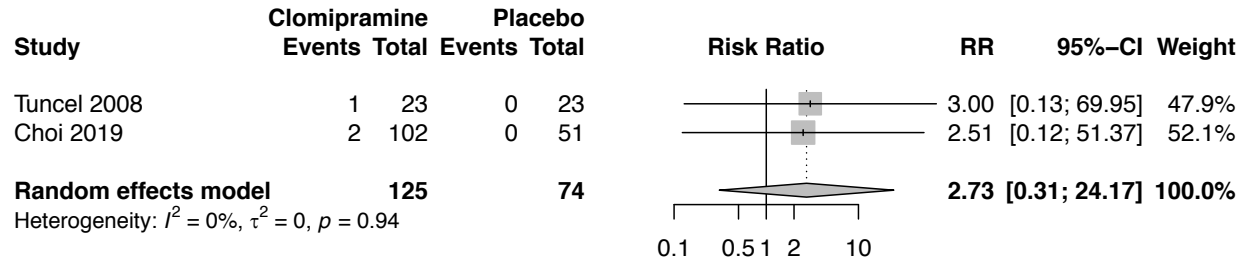


Figure S7
Forest plot of adverse events of renal and urinary system

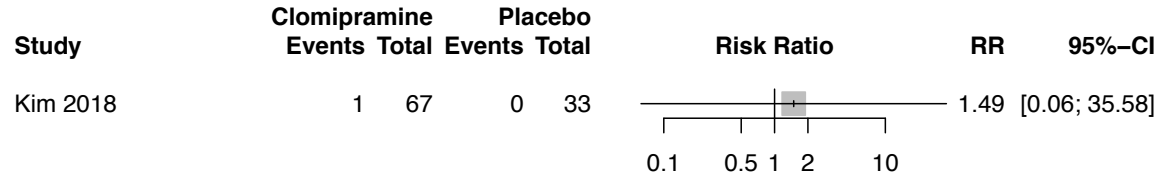


Figure S8
Forest plot of adverse events of reproductive system

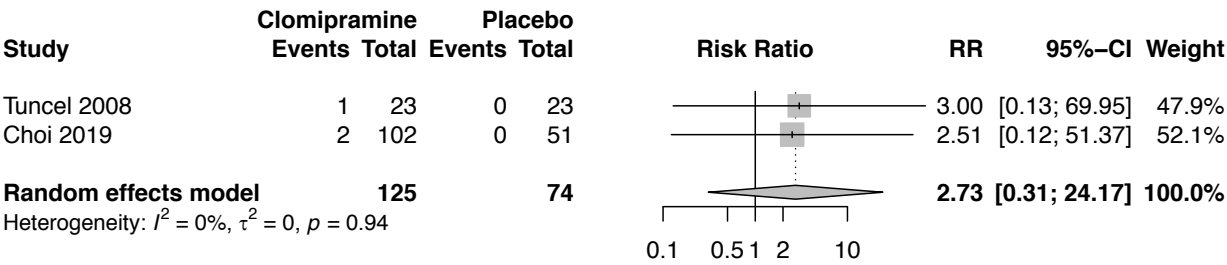


Figure S9
Forest plot of adverse events of dermatological system

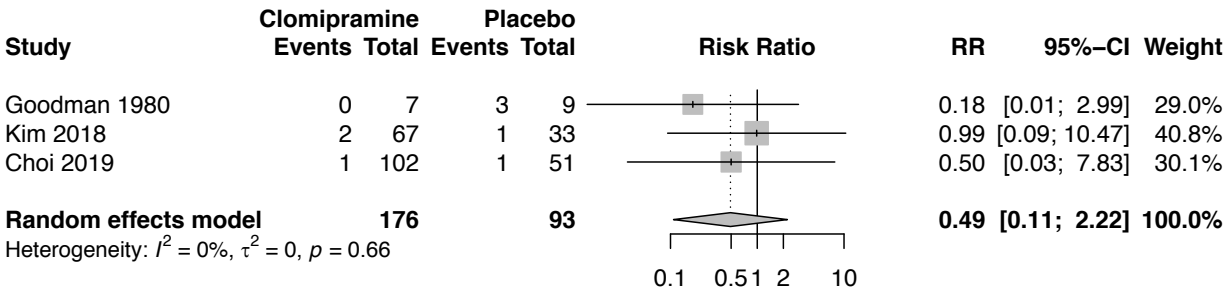
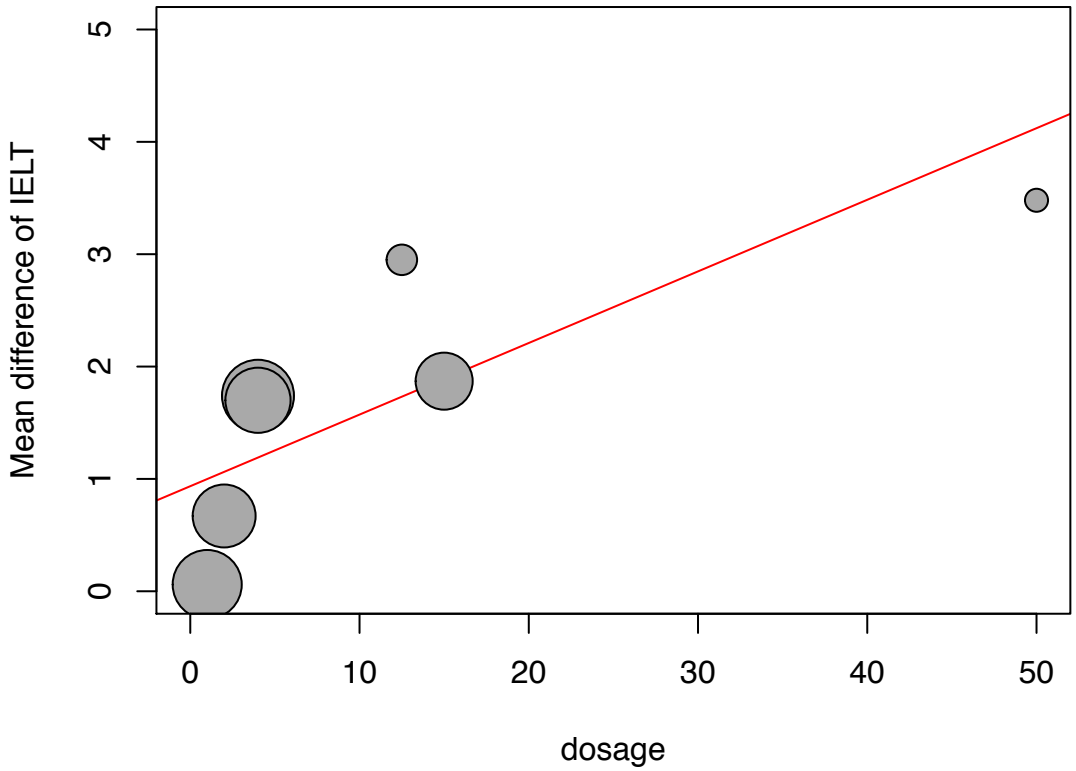


Figure S10 and Figure S11

Dose-dependent effects

Figure S10
Meta-regression analysis of IELT and clomipramine dosage

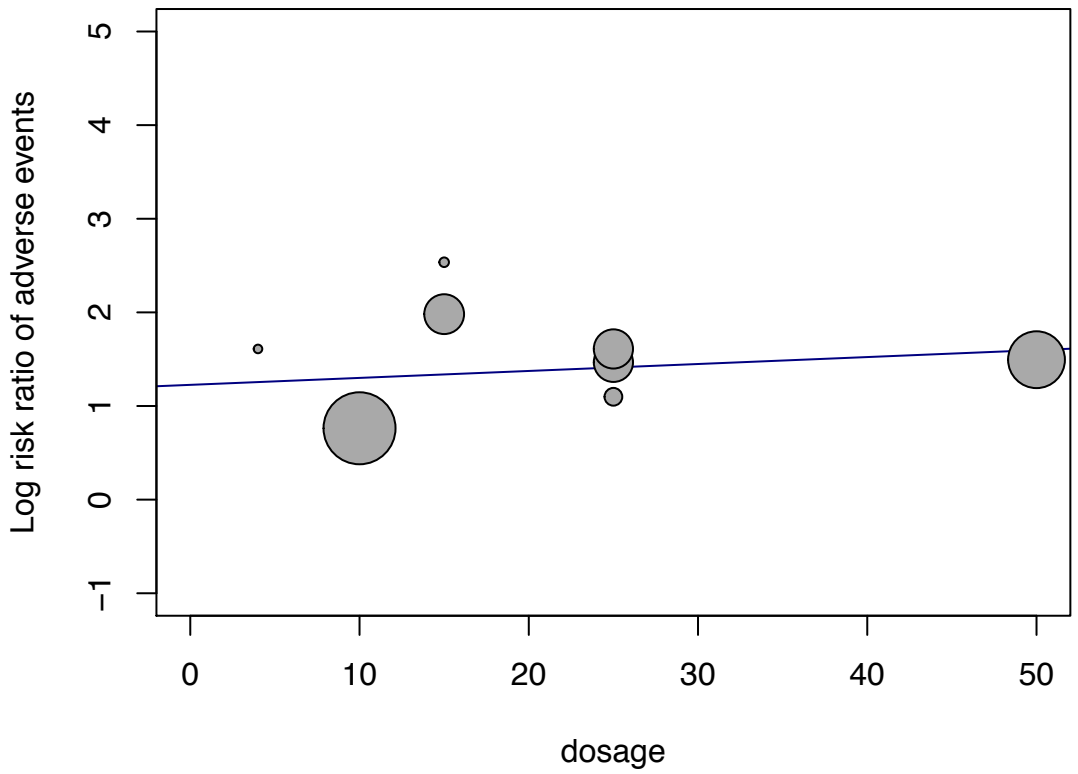


Slope intercept

	Point estimate	Standard Error	Lower limit	Upper limit	Z-value	P-value
Constant	0.9351	0.3603	0.2289	1.6413	2.5954	0.0094
Dosage	0.0637	0.0287	0.0074	0.1200	2.2181	0.0265

Figure S11

Meta-regression analysis of adverse events and clomipramine dosage



Slope intercept

	Point estimate	Standard Error	Lower limit	Upper limit	Z-value	P-value
Constant	1.2258	0.5110	0.2243	2.2273	2.3990	0.0164
Dosage	0.0074	0.0183	-0.0284	0.0433	0.4056	0.6850

Figure S12 to S15

Small study effect test of secondary outcomes

Figure S12
Forest plot of success rate

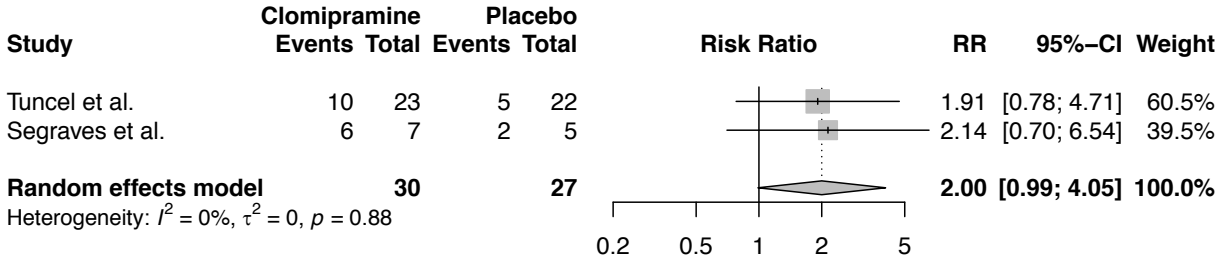
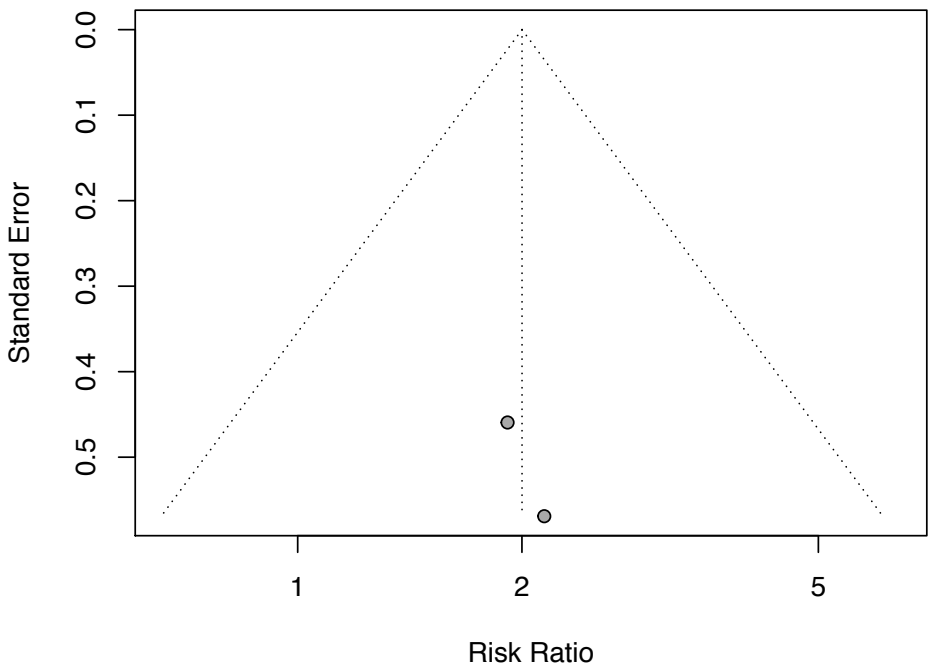


Figure S13
Small study effect test of success rate



Egger's test: not applicable

Figure S14
Forest plot of satisfaction rate

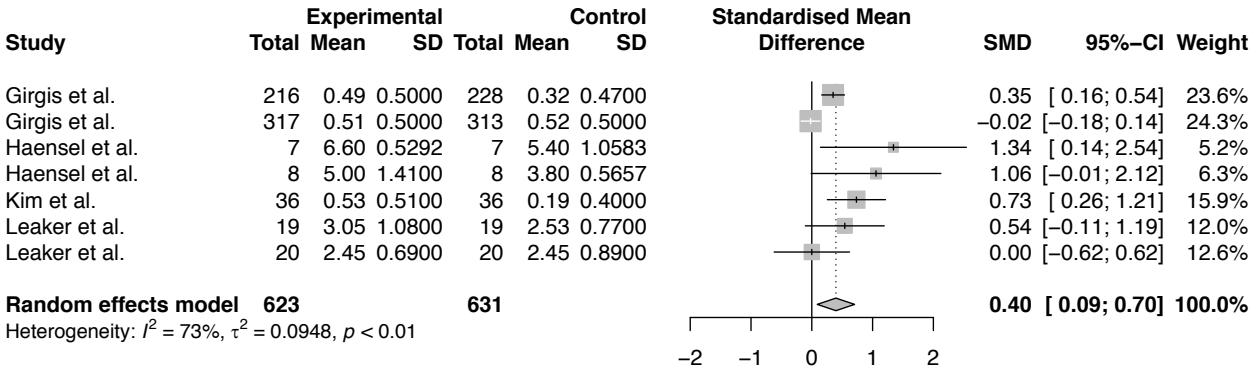
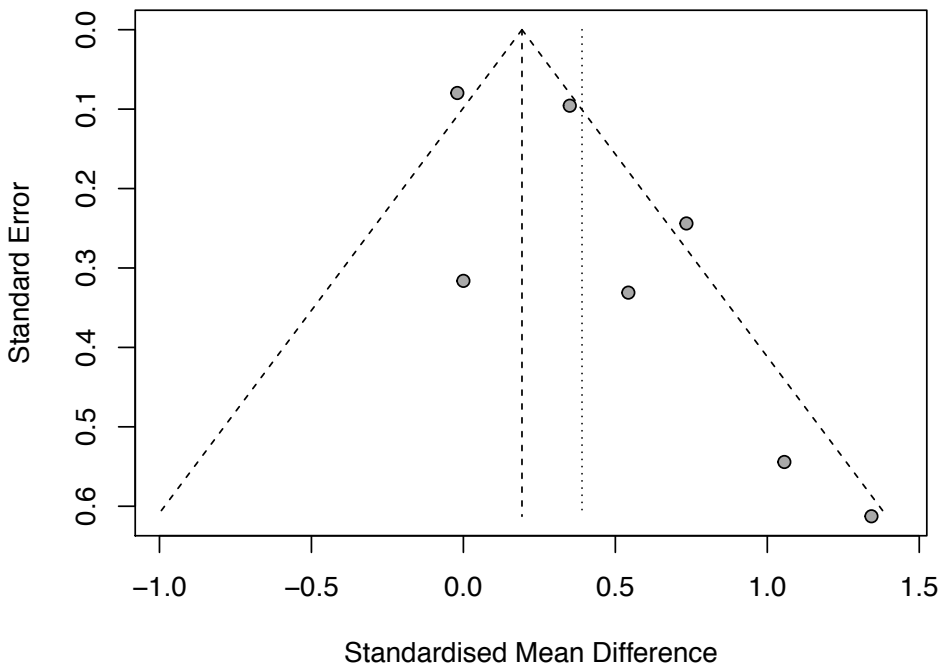


Figure S15
Small study effect test of satisfaction rate



Egger's test intercept

Intercept	95% confidence interval	t-value	P-value
1.944	-0.016 – 3.904	2.003	0.10156